

Influence of a Diet Very High in Vegetables, Fruit, and Fiber and Low in Fat on Prognosis Following Treatment for Breast Cancer

Context Evidence is lacking that a dietary pattern high in vegetables, fruit, and fiber and low in total fat can influence breast cancer recurrence or survival. **Objective** To assess whether a major increase in vegetable, fruit, and fiber intake and a decrease in dietary fat intake reduces the risk of recurrent and new primary breast cancer and all-cause mortality among women with previously treated early stage breast cancer. **Design, Setting, and Participants** Multi-institutional randomized controlled trial of dietary change in 3088 women previously treated for early stage breast cancer who were 18 to 70 years old at diagnosis. Women were enrolled between 1995 and 2000 and followed up through June 1, 2006.

Intervention The intervention group (n=1537) was randomly assigned to receive a telephone counseling program supplemented with cooking classes and newsletters that promoted daily targets of 5 vegetable servings plus 16 oz of vegetable juice; 3 fruit servings; 30 g of fiber; and 15% to 20% of energy intake from fat. The comparison group (n=1551) was provided with print materials describing the "5-A-Day" dietary guidelines. **Main Outcome Measures** Invasive breast cancer event (recurrence or new primary) or death from any cause. **Results** From comparable dietary patterns at baseline, a conservative imputation analysis showed that the intervention group achieved and maintained the following statistically significant differences vs the comparison group through 4 years: servings of vegetables, +65%; fruit, +25%; fiber, +30%, and energy intake from fat, -13%. Plasma carotenoid concentrations validated changes in fruit and vegetable intake. Throughout the study, women in both groups received similar clinical care. Over the mean 7.3-year follow-up, 256 women in the intervention group (16.7%) vs 262 in the comparison group (16.9%) experienced an invasive breast cancer event (adjusted hazard ratio, 0.96; 95% confidence interval, 0.80–1.14; P=.63), and 155 intervention group women (10.1%) vs 160 comparison group women (10.3%) died (adjusted hazard ratio, 0.91; 95% confidence interval, 0.72–1.15; P=.43). No significant interactions were observed between diet group and baseline demographics, characteristics of the original tumor, baseline dietary pattern, or breast cancer treatment. **Conclusion** Among survivors of early stage breast cancer, adoption of a diet that was very high in vegetables, fruit, and fiber and low in fat did not reduce additional breast cancer events or mortality during a 7.3-year follow-up period. **Trial Registration** clinicaltrials.gov Identifier: NCT00003787